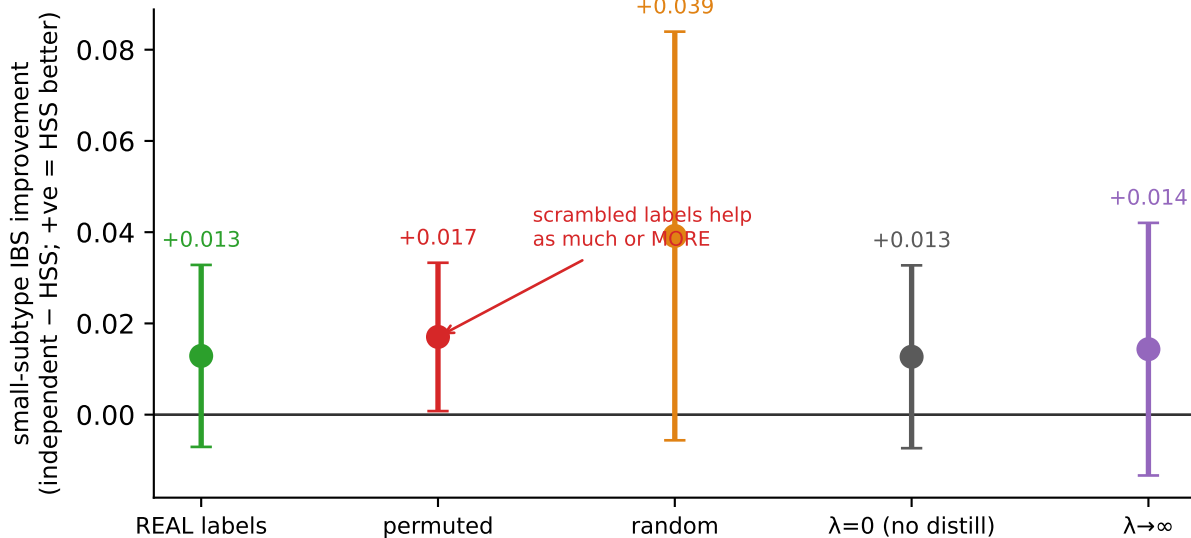


Subtype-aware NULL: REAL (+0.013) $\leq$ scrambled controls (max +0.039)
→ benefit is regularization, not cytogenetic biology (pre-registered STOP)



Small subtypes: del17p, del1p (sequencing-inferred, NOT FISH). OS technical validation only.